

0-1 Sequences

You are given a sequence, in the form of a string with characters '0', '1', and '?' only. Suppose there are k '?'s. Then there are 2^k ways to replace each '?' by a '0' or a '1', giving 2^k different 0-1 sequences (0-1 sequences are sequences with only zeroes and ones).

For each 0-1 sequence, define its number of inversions as the minimum number of adjacent swaps required to sort the sequence in non-decreasing order. In this problem, the sequence is sorted in non-decreasing order precisely when all the zeroes occur before all the ones. For example, the sequence 11010 has 5 inversions. We can sort it by the following moves: 11010 $\rightarrow$ 11001 $\rightarrow$ 10101 $\rightarrow$ 01101 $\rightarrow$ 01011 $\rightarrow$ 00111.

Find the sum of the number of inversions of the 2^k sequences, modulo 1 000 000 007 ($10^9 + 7$).

Input

The first and only line of input contains the input string, consisting of characters '0', '1', and '?' only, and the input string has between 1 to 500 000 characters, inclusive.

Output

Output an integer indicating the aforementioned number of inversions modulo 1 000 000 007.

Sample Input 1

?0?



Sample Output 1

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